

**A Multifaceted Exploration of Spatial
Openness in Rental Housing: Big Data
Analysis Across Tokyo's 23 Wards**

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Abstract

Understanding spatial openness in residential housing is important for enhancing living quality and guiding architectural design. This study proposes a multifaceted evaluation method of spatial openness in rental housing by leveraging large-scale data from Tokyo's 23 wards. Visibility Graph Analysis (VGA) is applied to floor plan images to assess two-dimensional spatial connectivity. In addition, semantic segmentation is performed on interior photos to capture three-dimensional openness features such as window ratios and ceiling visibility. The study investigates how openness characteristics vary by geographic location and construction year, identifying spatial and temporal patterns. Finally, it explores the relationship between these openness features and subjective housing preferences, providing insights into how spatial openness influences perceived residential quality.

Acknowledgements

This is a good place to acknowledges those who helped you along the way.

Also please note how LaTeX has put this on a new page as it is a new chapter. No command (apart from starting a new chapter) was needed to do this.

Also note the lack of number on the title, and contrast this to proceeding chapters, the abstract and acknowledgements should be unnumbered, the proceeding chapters should be numbered.

Chapter 1

Introduction

1.1 Research Background and Motivation

Understanding spatial openness in residential housing is fundamental to enhancing living quality and guiding architectural design practices. In urban environments, particularly in densely populated cities like Tokyo, the perception of spatial openness significantly impacts residents' psychological well-being, comfort, and overall satisfaction with their living spaces. Traditional architectural evaluation methods have long relied on basic quantitative metrics such as floor area, room count, and ceiling height to assess spatial quality. However, these conventional indicators fail to capture the nuanced spatial qualities that contribute to the subjective experience of openness in residential environments.

The concept of spatial openness extends beyond mere physical dimensions to encompass the complex interplay between visual connectivity, visibility of interior elements, and so on. In the context of rental housing markets, where prospective tenants make decisions based on limited visual information from online platforms, understanding how spatial openness translates through digital representations becomes increasingly important. The proliferation of online rental platforms has created unprecedented opportunities to analyze large-scale housing data, including interior imagery and floor plans, which can provide insights into

spatial characteristics that were previously difficult to quantify systematically.

Tokyo’s 23 special wards represent a unique data playground for studying spatial openness due to the city’s distinctive urban morphology, diverse architectural styles spanning multiple decades, and the comprehensive digitization of rental housing information, which has always been one of the most populated cities and rental market in the world.

1.2 Literature Review and Research Gaps

1.2.1 Spatial Openness in Architectural Research

The study of spatial openness has been approached from various perspectives within architectural and environmental psychology research. Traditional spatial analysis methods have focused on geometric properties and numerical metrics of building attributes. Space syntax theory, developed by Hillier and Hanson, introduced systematic approaches to understanding spatial relationships through graph-based analysis of architectural layouts [1]. Visibility Graph Analysis (VGA), as an extension of space syntax principles, provides a quantitative framework for measuring visual connectivity and spatial integration within architectural spaces.

Previous research has demonstrated that spatial openness influences occupant behavior, movement patterns, and psychological responses to built environments. Studies in environmental psychology have shown that perceived openness correlates with reduced stress levels, improved cognitive performance, and enhanced overall well-being. However, most existing research has been limited to small-scale case studies or controlled experimental settings, lacking the scale and diversity necessary to understand spatial openness patterns across entire urban regions.

1.2.2 Computer Vision and Spatial Analysis

Recent advances in computer vision and deep learning have opened new possibilities for automated spatial analysis. Semantic segmentation techniques enable the extraction of architectural elements from images, allowing for systematic analysis of interior compositions. Similarly, these deep-learning-based models have been successfully applied to recognize and classify various architectural features in floor plans, including walls, windows, floors, and ceilings. These technological developments create opportunities to process large-scale image datasets that would be impractical to analyze manually.

The integration of 2D image analysis with 3D spatial analysis recently draws more attentions in architectural research. While 2D analysis can capture visual qualities and material compositions, 3D analysis provides insights into spatial relationships and configurational properties. The combination of these approaches offers a more comprehensive understanding of spatial openness than either method alone.

1.2.3 Big Data Opportunities and Research Gaps

The emergence of big data analytics in housing research has revealed new patterns in residential preferences, market dynamics, and architectural trends. Online rental platforms generate vast amounts of data, including property specifications, pricing information, and visual documentation through interior images and floor plans. This data richness enables researchers to investigate questions about spatial quality and market valuation at unprecedented scales, creating opportunities that were previously impossible due to data limitations.

However, the potential for using big data to understand subjective spatial experiences remains largely unexplored, particularly in the context of rental housing markets that traditionally rely on rule-based tabular data without in-depth understanding of spatial qualities. While studies using conventional tabular data can achieve large scale, spatial openness research involving image analysis or human-operated software assessments has been limited to small samples, preventing generalization to broader urban contexts. Additionally, the tools

and methodologies for systematically processing and analyzing large-scale visual and spatial data in housing research are still developing, creating both opportunities and challenges for comprehensive spatial analysis.

1.3 Primary Research Objectives

This research aims to address the identified gaps through the following primary objectives:

1. **Develop a Comprehensive Methodology:** Create a multifaceted evaluation method for spatial openness that combines automated extraction of visual elements from interior images with floorplan semantic segmentation for VGA simulations, enabling large-scale analysis previously limited by manual assessment constraints.
2. **Large-Scale Analysis:** Conduct systematic analysis of spatial openness using large-scale rental housing data from Tokyo’s 23 wards.
3. **Temporal and Geographic Investigation:** Examine how spatial openness varies across construction periods and geographic regions within Tokyo’s 23 wards.
4. **Feature Dependency Analysis:** Investigate relationships between 2D visual elements and 3D spatial connectivity measures to identify key interactions contributing to spatial openness perception.
5. **Market Correlation Analysis:** Explore the relationships between spatial openness features, impression evaluations and rental costs.

1.4 Structure of the Thesis

This thesis is organized into six main chapters, each addressing specific aspects of spatial openness analysis:

Chapter 1: Introduction provides the research background, literature review, objectives, and methodology overview.

Chapter 2: Methodology presents detailed descriptions of the Lifull dataset, openness quantification methods including interior image-based 3D perception analysis and floor plan-based 2D VGA analysis, and statistical analysis frameworks.

Chapter 3: Results presents the primary findings, including temporal analysis of openness characteristics across construction decades and geographic distribution analysis across Tokyo’s 23 wards.

Chapter 4: Comprehensive Openness Analysis provides in-depth analysis including regression analysis against rent costs and detailed analysis of openness features, with image examples and comparison between 2D and 3D openness measures.

Chapter 5: Discussion interprets the results, discusses implications for architectural design and urban planning, addresses limitations, and suggests future research directions.

Chapter 6: Conclusion summarizes the key findings, contributions, and practical implications of the research.

The thesis concludes with comprehensive appendices containing detailed technical specifications, additional statistical analyses, and supplementary materials that support the main findings.

This research contributes to the growing field of computational architecture and urban analytics by providing a systematic framework for understanding spatial openness in residential environments. The findings have implications for architectural design, urban planning, and real estate market analysis, offering data-driven insights that can inform more responsive and human-centered housing design practices.

Chapter 2

Data Source and Preprocessing

2.1 Overview of the LIFULL Real Estate Dataset

2.2 Definition of Openness

2.3 Data Filtering and Cleaning

2.4 Generation of Subjective Impression Scores (Introduction of the Impression Evaluation Model)

Chapter 3

Quantifying the Openness of Residential Spaces

Temporary placeholder citation to satisfy BibTeX: [2]

3.1 Floor Plan Processing and Visibility Graph Analysis (VGA)

3.2 Openness Feature Extraction from Interior Images

- Estimating Window Proportions
- Segmenting Ceilings, Walls, Doors, etc.

3.3 Matching Strategy for Floor Plans and Interior Images

3.4 Visualization Examples of Openness Features

Chapter 4

Influence of Age and Geographic Factors

- 4.1 Impact of Building Age on Openness-Impression Relationship
- 4.2 Comparative Analysis Across Geographic Areas
- 4.3 Spatial Visualization and Cluster Analysis

Chapter 5

Correlation Analysis Between Subjective Impressions and Openness

5.1 Impression Score Outputs and Interpretation

5.2 Correlation Between Openness Features and Subjective Impressions

5.3 Case Studies and Visual Comparisons

Chapter 6

Conclusion and Future Work

6.1 Summary of Findings and Contributions

6.2 Limitations and Sources of Error

6.3 Future Research Directions

Appendix A

An example of an appendix

This is what an appendix looks like!

Appendix B

Session Protocol

Zigzag Session Protocol

Samuel Meijer

March 2019

1 Introduction and Motivation

In order to understand how to steer a boat, it is necessary to understand how the boat responds to rudder inputs. The methods used in this protocol to measure the response of the boat are adapted from the methods presented in the work of Abkowitz¹ and Lewis².

2 Description

The goal of this session is to perform the zigzag maneuver (also known as *Kempf Overshoot* or “Z” maneuver³) to derive a relationship between the rudder angle, δ , and the rate of rotation of the boat, $\dot{\theta}$. This involves steering the boat in a controlled zigzag pattern.

3 Procedure

The maneuver is conducted as follows:
while rowing,

1. the boat is set travelling straight (rudder angle, $\delta = 0$)
2. the rudder is turned to a set angle ($\delta = \delta_1$) for a set period of time, τ (until the change of rate of rotation, $\dot{\theta}$, is equal to zero [$\dot{\theta} = 0$])
3. the rudder is turned to a set angle in the opposite direction ($\delta = -\delta_1$)

Steps 2 and 3 are repeated a number of times the same rudder angles, δ_1 and the same length of time for the rate of rotation to approach zero, τ .

For this session, the rudder position will be controlled **automatically** by a servo running a programmed course. In this case, $\tau = 15s$ and $\delta_1 = -\delta_2 = 20^\circ$.

¹Martin A. Abkowitz. *Measurement of Ship Hydrodynamic Coefficients in Maneuvering From Simple Trials During Regular Operations*. M.I.T. Department of Ocean Engineering, Cambridge, November 1984.

²Edward V. Lewis. *Principles of naval architecture*. Society of Naval Architects and Marine Engineers, Jersey City, 2nd revision (3rd ed.) 1988.

³Ibid.

Appendix C

Software

```
1 #include <Wire.h>
2 #include <Adafruit_Sensor.h>
3 #include <Adafruit_BNO055.h>
4 #include <utility/imumaths.h>
5 #include <Adafruit_GPS.h>
6 #include <SPI.h>
7 #include <SD.h>
8 #include <Servo.h>
9 #include <math.h>
10 #include "Filter.h"
11
12 /* This driver reads raw data from the BNO055, Potentiometer and
    GPS
13
14 Connections for ADALOGGER
15 =====
16
```

```

17 Potentiometer
18 _____
19 Connect one side to VCC
20 Connect other to common ground
21 Connect Middle to A0
22
23 GPS
24 _____
25 Connect VIN to VCC
26 Connect GROUND to common ground
27 Connect GPS TX to RX1 (D0)
28 Connect GPS RX to TX1 (D1)
29
30 */
31
32 /*****/
33 /*BN0055 (orientation) setup, variables and def'n*/
34 /*****/
35 // Sample delay
36 const float SAMPLERATE = 10;
37 // Setup for the differentiation variables
38 double psierror = 0;
39 double dt = 0;
40 double dpsierror_dt = 0;
41 double previouspsierror = 0;

```

Bibliography

- [1] B. Hillier, J. Hanson, and H. Graham, “Ideas are in things: an application of the space syntax method to discovering house genotypes,” *Environment and Planning B: planning and design*, vol. 14, no. 4, pp. 363–385, 1987.
- [2] D. Author, “Placeholder reference,” *Placeholder Journal*, 2024. This is a temporary placeholder reference.